

General Sensory Systems

Principles and Practices of Osteopathic Medicine I

- *Jennifer Xie, Ph.D.*
- Associate Professor
- Biomedical and Anatomical Sciences
- Arkansas, Wilson Hall, Rm # 144J

Email: Jennifer.xie@nyit.edu



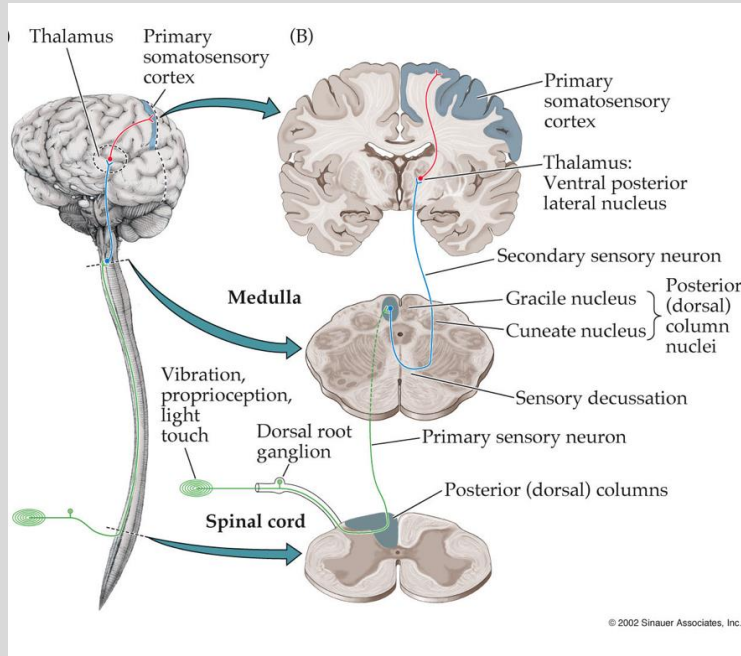
Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

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- Identify **somatosensory receptors** and their encoding.
- Describe the transmission of **touch and proprioception** to the cerebral cortex.
- Describe the transmission of **pain and temperature** information to the cerebral cortex.
- Understand the relation between somatosensory lesions and clinical symptoms.



See **detailed objectives** in CPG.

PLEASE READ THE Scholar Rx Brick!

Session Objectives Mapped to COMLEX Blueprint

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- **Dimension 2**
 - **Clinical presentation 4.** Patient presentations related to the nervous system and mental health
 - 4.11 Sensory disturbances and pain
 - tactile disturbances, including sensory loss, numbness, vibration/temperature/proprioception loss, tingling, and paresthesia
 - pain, chronic nonmalignant
 - pain, neuropathic, nociceptive, mixed, sympathetic
 - nerve-, muscle-, and pain-related syndromes, including complex regional pain syndrome, post-herpetic neuralgia, meralgia paresthetica

Session Objectives Mapped to USMLE Blueprint

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V. Nervous System & Special Senses

A. Normal Processes

2. Organ structure and function

iv. Sensory systems

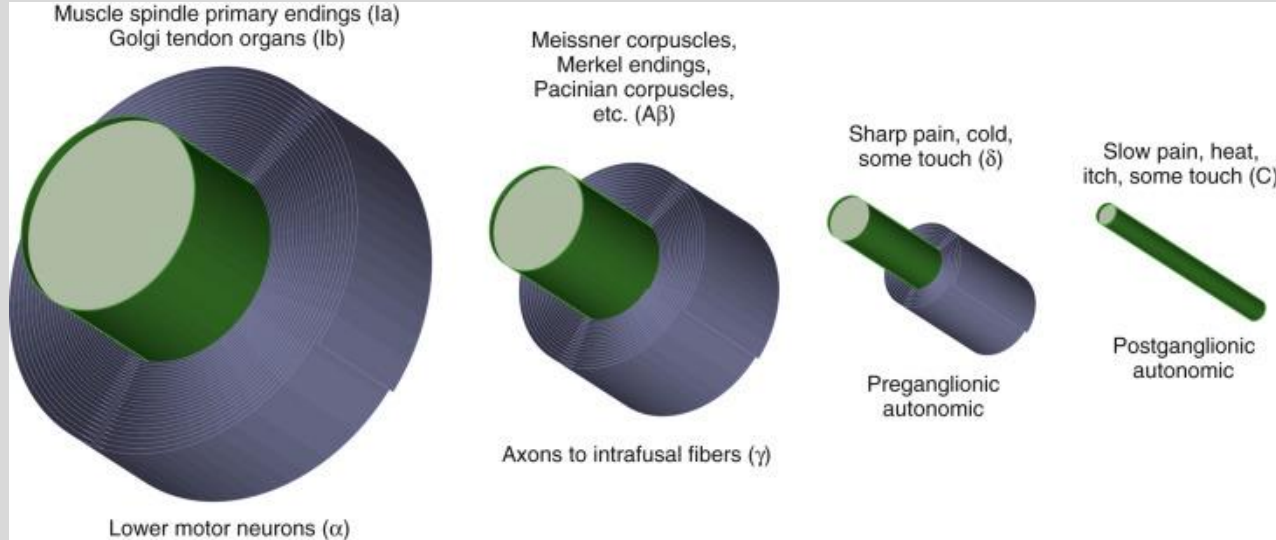
a. general sensory modalities, including sharp, dull, temperature, vibratory, and proprioception

B. Abnormal Processes: Health and Health Maintenance, Screening, Diagnosis, Management, Risks, Prognosis

5. Cranial and peripheral nerve disorders

iii. Neurologic pain syndromes: complex regional pain syndrome (reflex sympathetic dystrophy, causalgia); fibromyalgia; postherpetic neuralgia; phantom limb pain/syndrome; thalamic pain syndrome; trigeminal neuralgia

Different size, sheath, and speed of afferent fibers



Conduction velocity increases with diameter & myelination
Range from 0.5 m/s – 60 m/s*

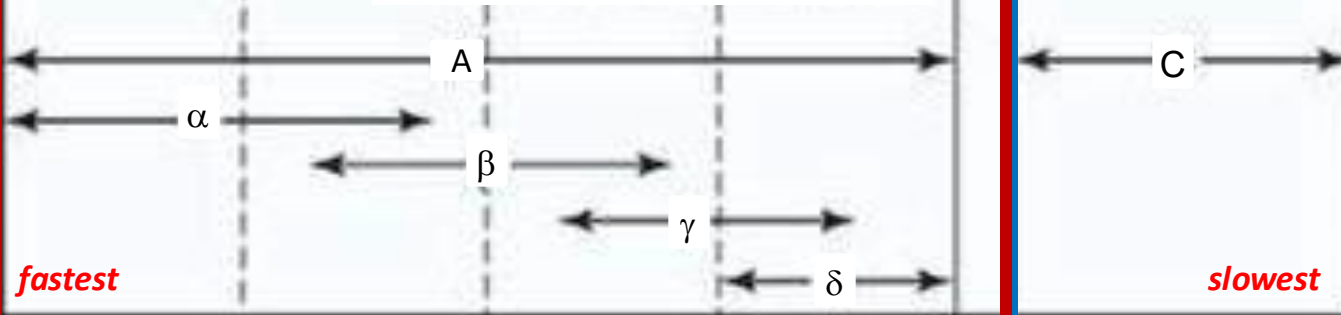
*textbooks often report a max of 120 m/s which comes from cats, not people.

Classification of peripheral fibers

Myelinated

Unmyelinated

Based on conduction velocity



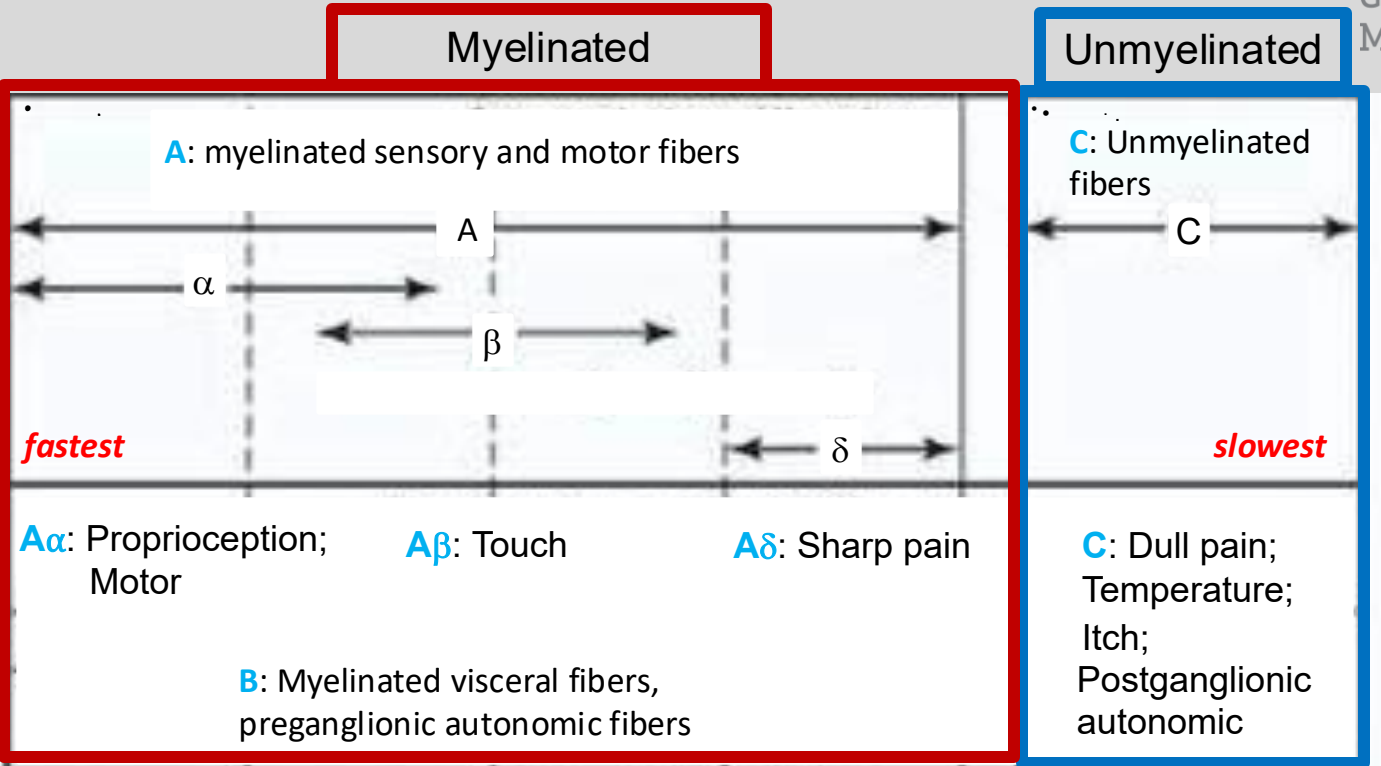
Based on axonal diameters
(no need to remember)



Reasonable Simplification

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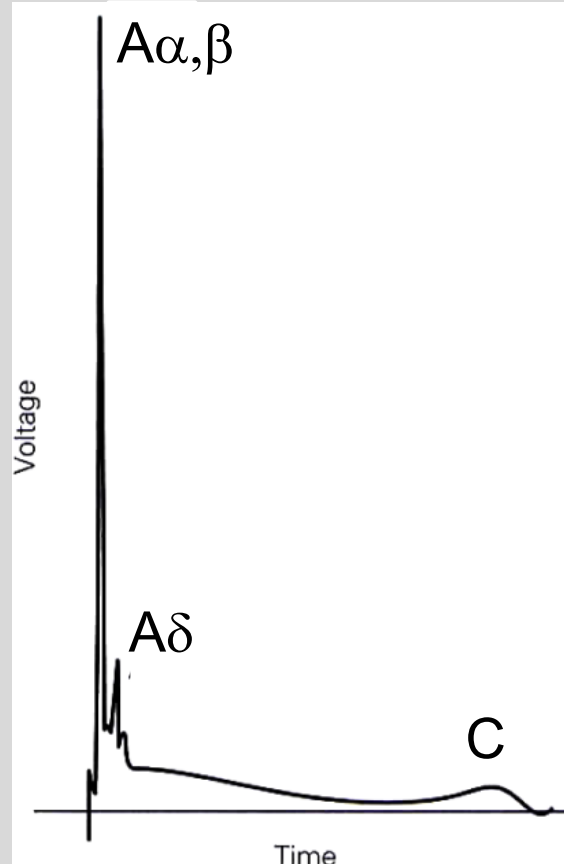


Conduction Speed

A strong electrical stimulus of the skin will evoke a compound action-potential in a nerve fiber.

Its complex waveform reflects the summed response of individual axons having different diameters or myelination

Note the delayed response by C fibers



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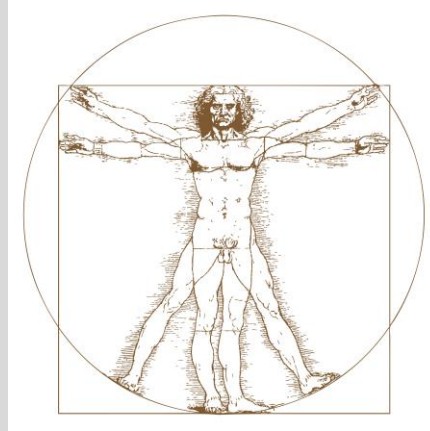
Diversity of receptors for somatosensation

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1 Rod
3 Cones



1 Hair cell

5 touch
2 proprioception
1 force, 1 joint, 1 hair
> 2 temperature
> 2 pain/itch/ache
visceral
(plus more not mentioned &
being discovered)

Categories of receptors for somatosensation

Based on the **source** of stimuli

• **Interoceptors:**

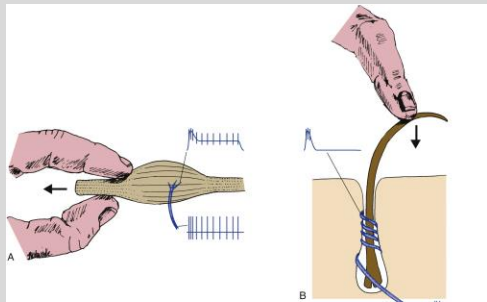
- Internal stimuli within the body, e.g., baroreceptors, pH sensors

• **Exteroceptors**

- External stimuli outside the body, e.g., skin touch, heat sensors

• **Proprioceptors**

- Position changes of the body parts, e.g., receptors at joints



Based on the **types** of stimuli

• **Chemoreceptors**

- Smell, taste, pH, etc.

• **Photoreceptors**

- Visual signals

• **Thermoreceptors**

- Temperature sensors

• **Mechanoreceptors**

- Detecting physical deformation

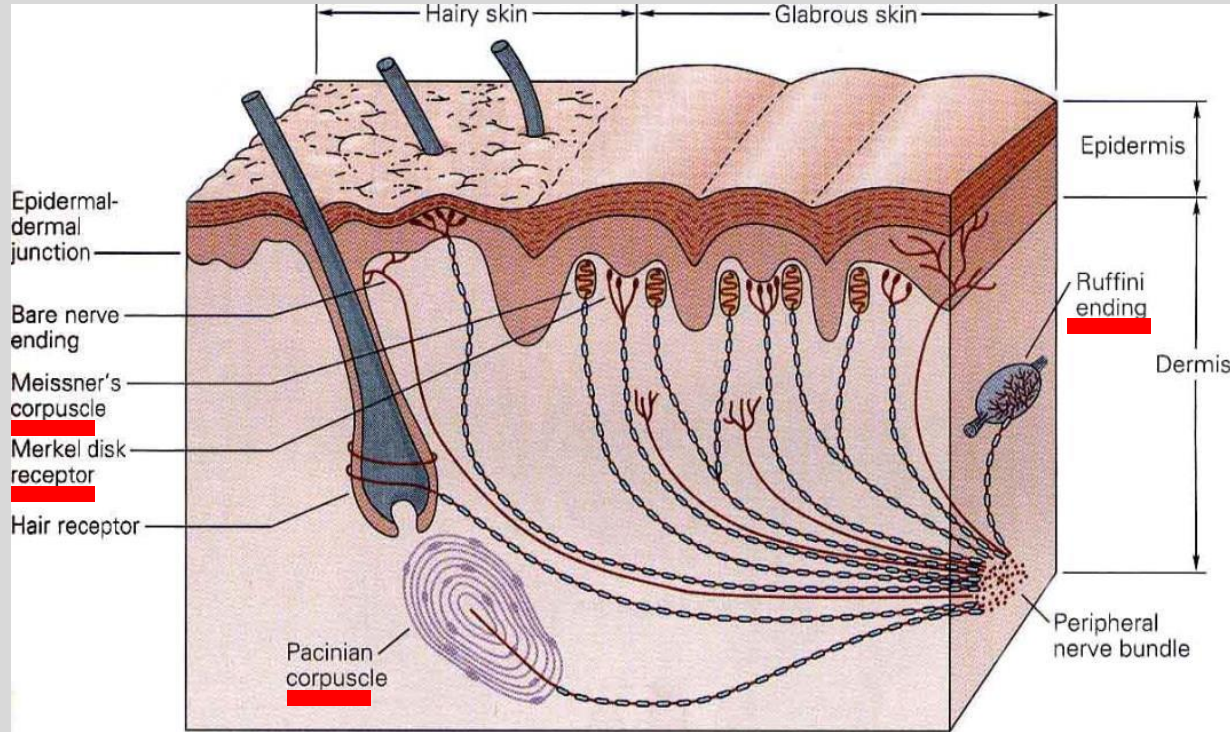
• **Nociceptors** (Latin “*nocere*” = to hurt)

- Noxious stimuli: mechanical, thermal, chemical, etc.

Skin/cutis is richly embedded with touch sensors

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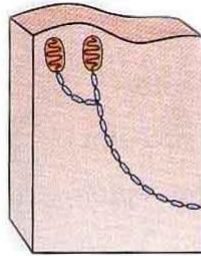


These are “exteroceptors”, also “mechanoreceptors” (type of receptors).

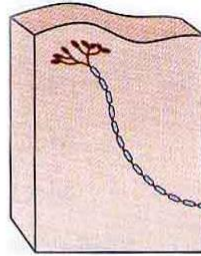
Source: Kandell and Schwartz, Principles of Neuroscience

Touch receptors have different sensitivity

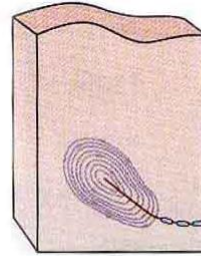
Receptors



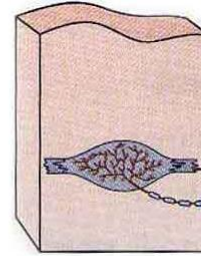
Meissner's
corpuscle



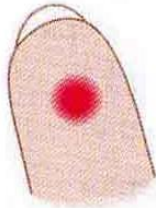
Merkel
cells



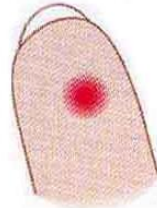
Pacinian
corpuscle



Ruffini
endings



Small RF



Small RF



Large RF



Large RF

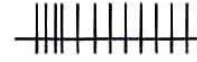
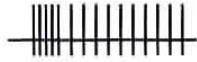
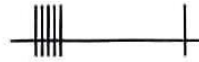
Fast Adapting

Slow Adapting

Fast Adapting

Slow Adapting

Neural
spike train



Stimulus



Touch

Vibration, pressure

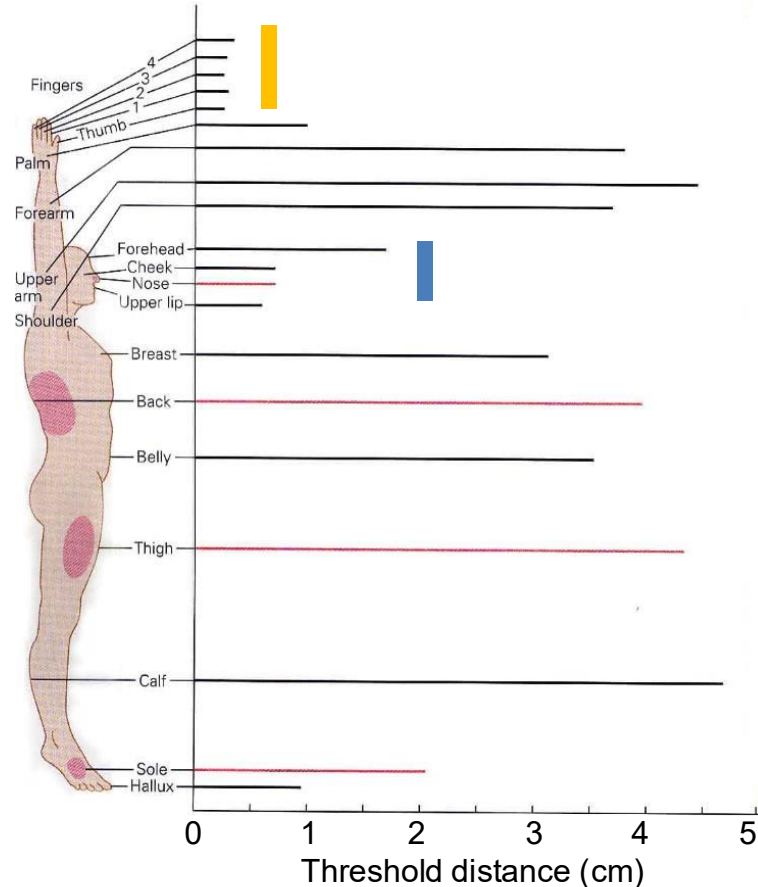
Touch sensitivity varies across body surface

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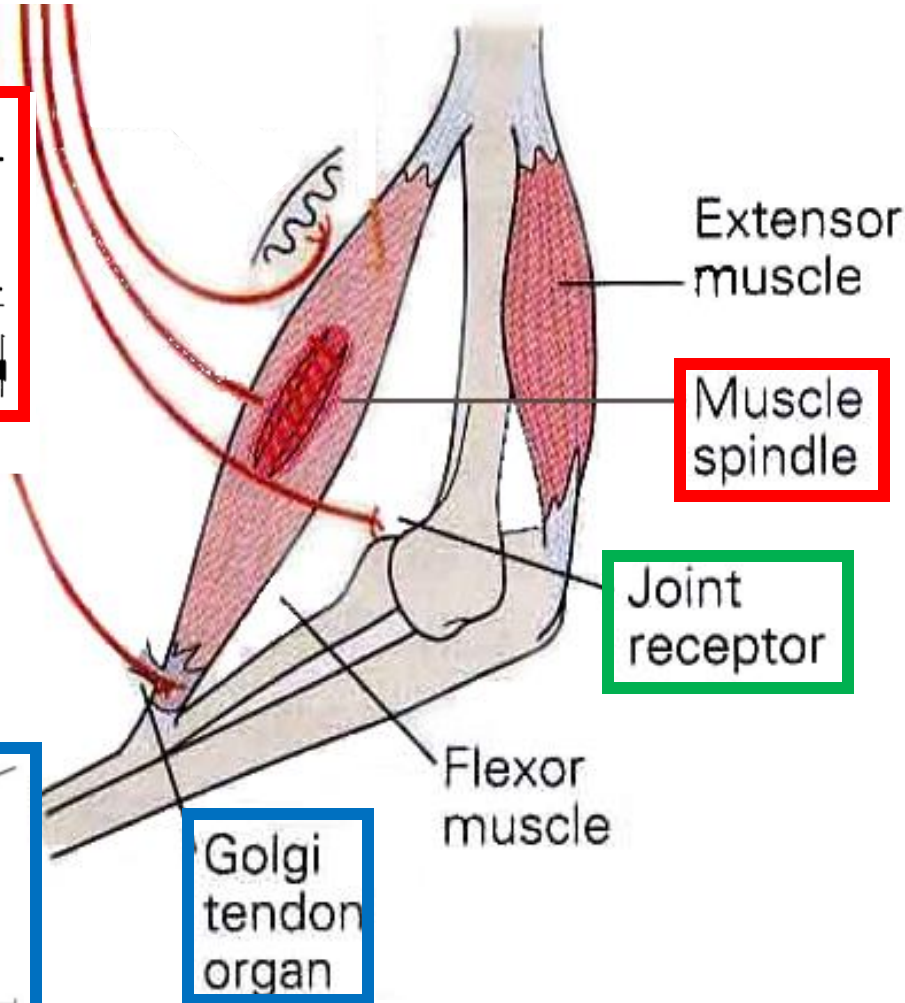
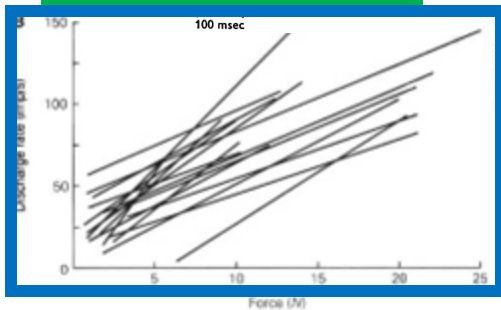
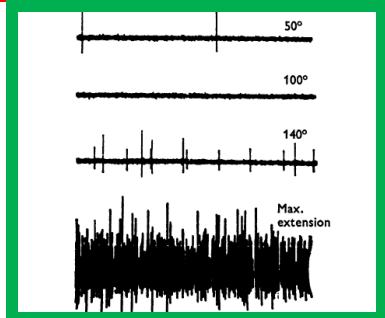
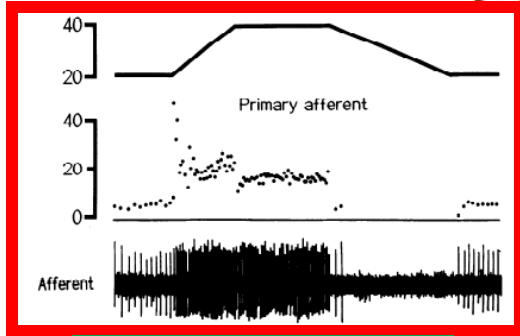
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- Reflects **density** of receptors
- Packed highest where there is most important physical interaction: **hands** and **face**

Use two point-discrimination to assess: is it “one” or “two”?



Proprioceptors



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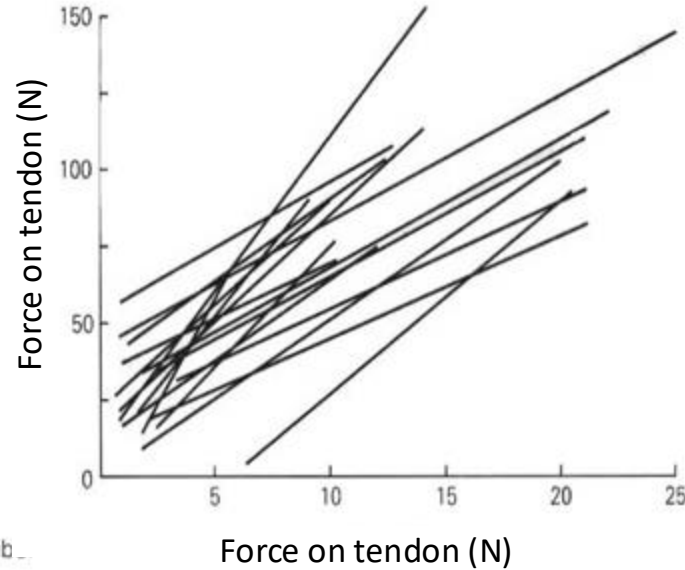
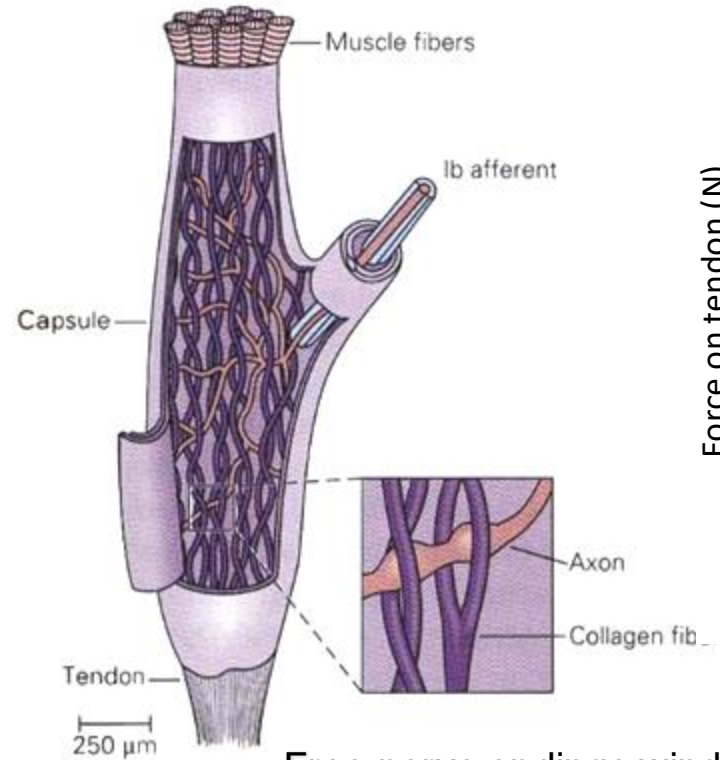
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“**proprioceptors**” convey information about the state of the musculo-skeletal apparatus.

Golgi tendon organs sense force at the tendon

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Free nerve endings wind through collagen

Pulling force at the junction (from external load or active muscle)
bends the free nerve endings, opens the ion channels,
and depolarizes the membrane

Joint receptors are sensitive to capsule strain

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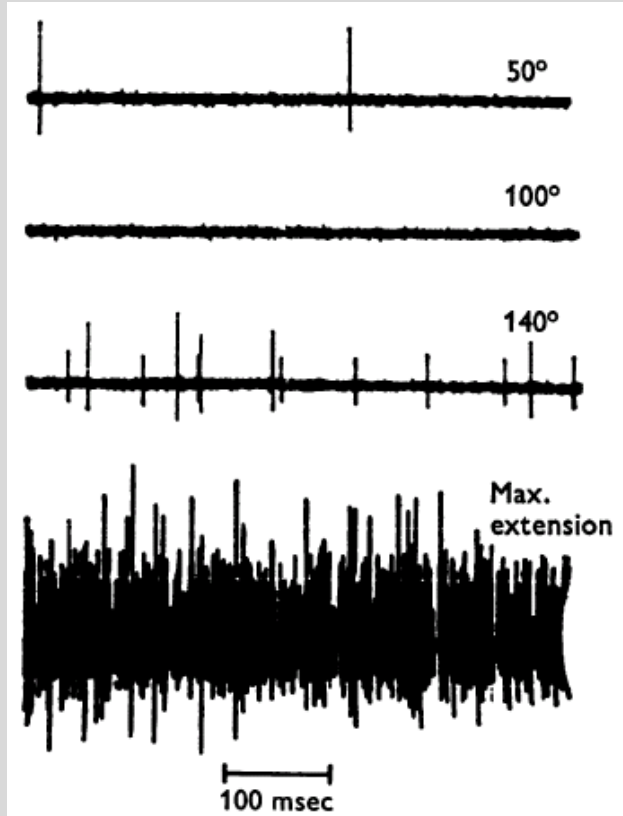
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Free nerve endings embedded
within the joint capsule

Insensitive to limb position within
its normal range of motion

Detects limb position at
extremes and/or inappropriate
directions

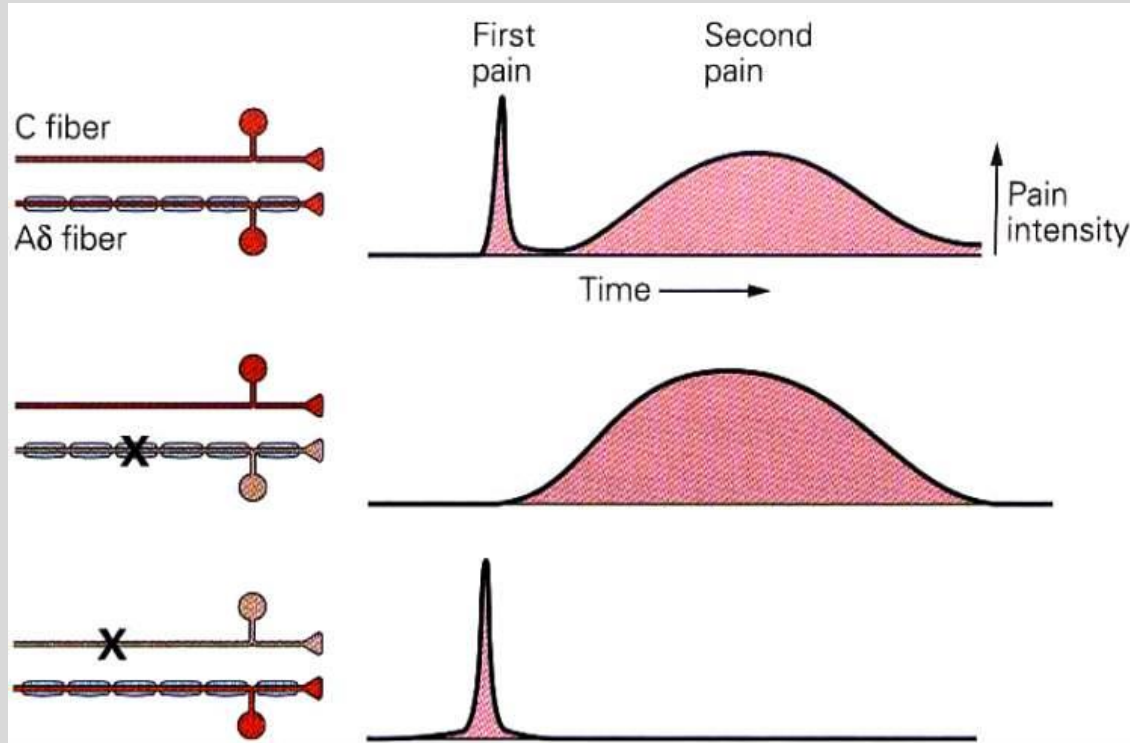
For example, they are recruited
during valgus/varus motion of
the knee



Nociceptors for pain sensation

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Aδ fibers - thermal, mechanical nociceptors

C fibers - polymodal nociceptors including chemosensation

Ascending modulation of pain

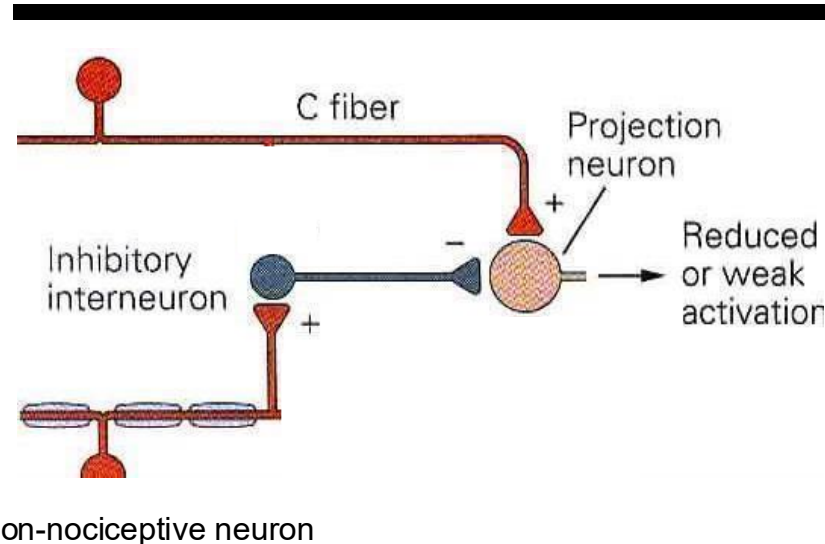
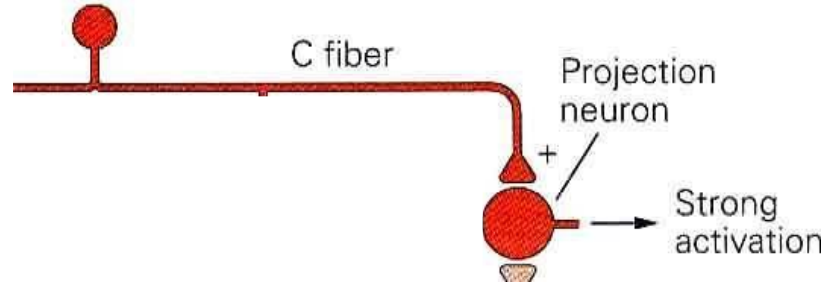
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Gate Theory

Low-threshold tactile afferents inhibit nociceptive pathway, i.e. "closes a gate".

The projection neuron response is weakened when the non-pain activity occurs during C fiber activity.

Explains why rubbing the area around your injury makes it feel better.



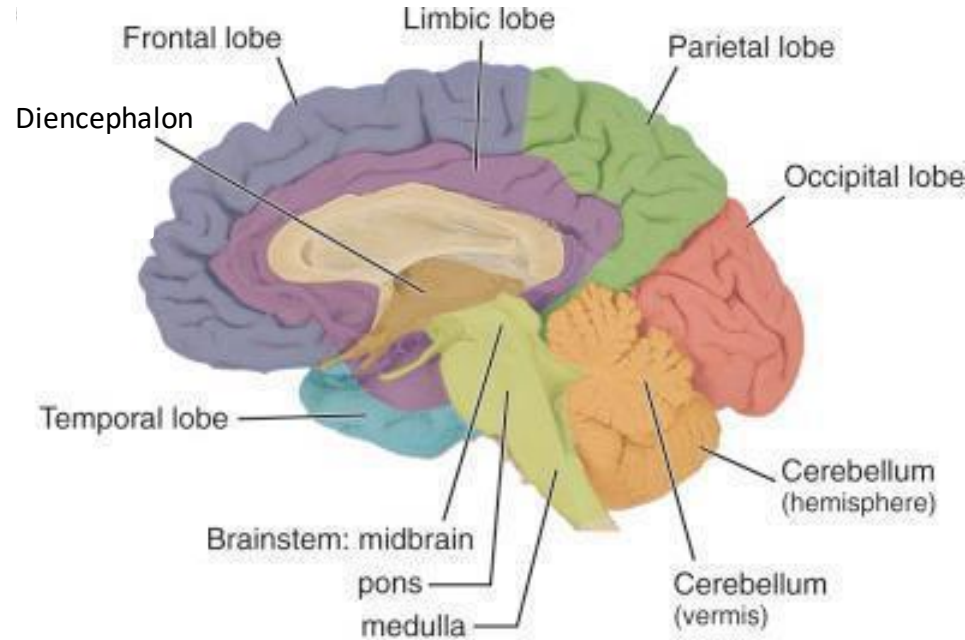
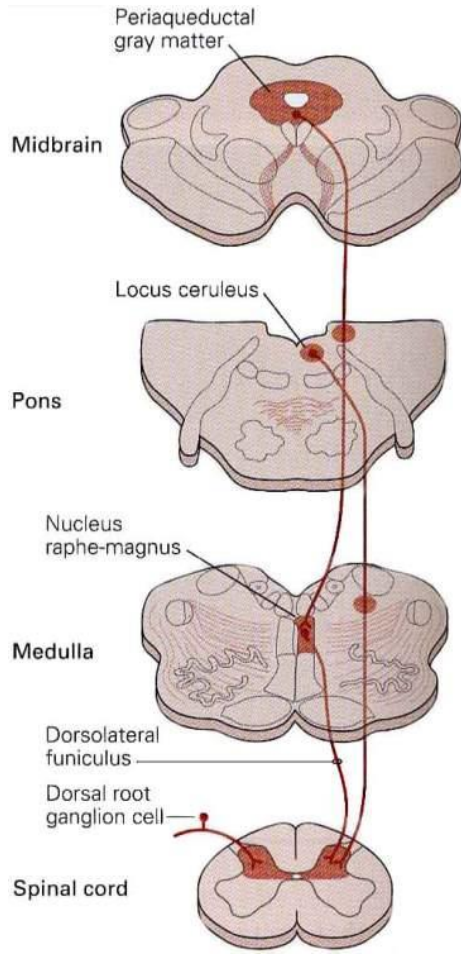
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Descending modulation of pain

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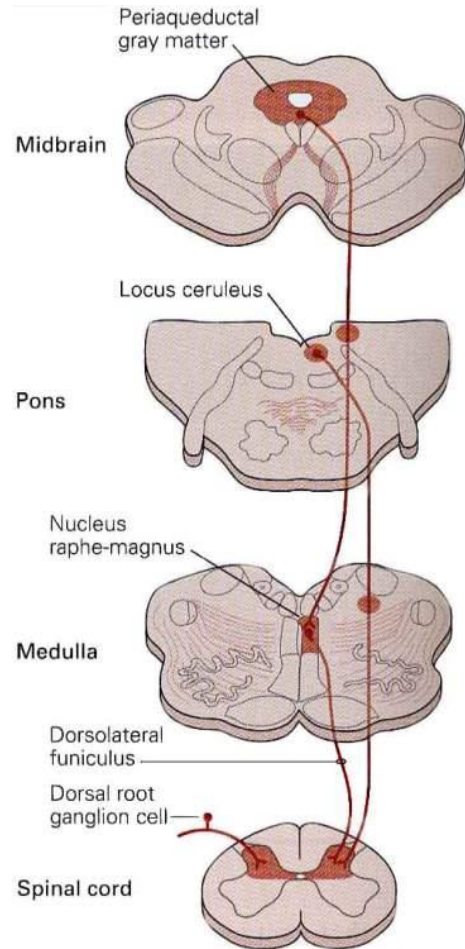
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Descending modulation of pain

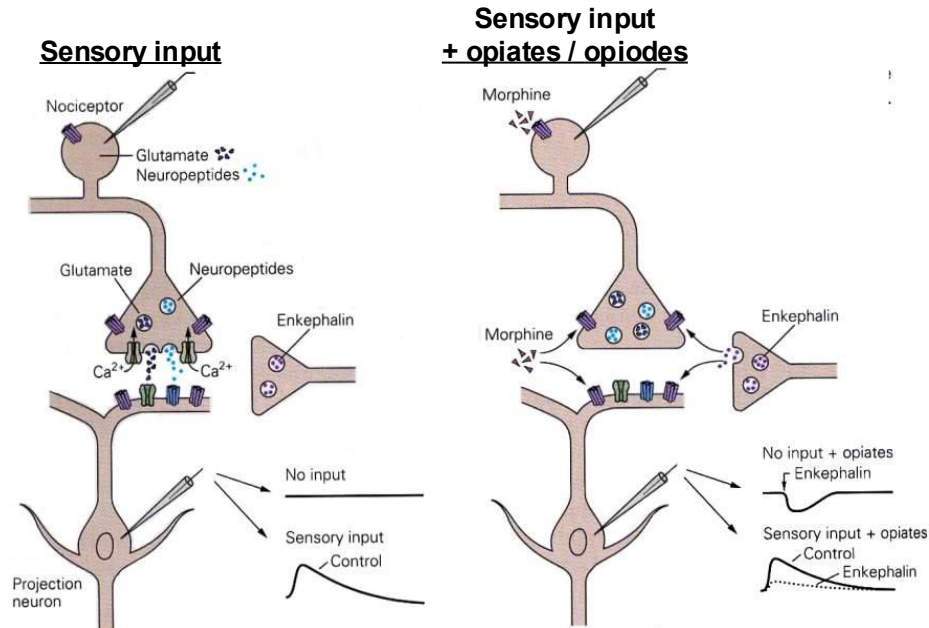
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Stimulation of periaqueductal gray matter excites locus coeruleus (NE) and raphe nucleus (5HT, AKA serotonin). These project to the dorsal horn (substantia gelatinosa) and activate enkephalin-containing interneurons:

- 1) Prevent calcium influx in the presynaptic pain neuron
- 2) Which reduce transmitter release
- 3) And reduces postsynaptic depolarization and activity.



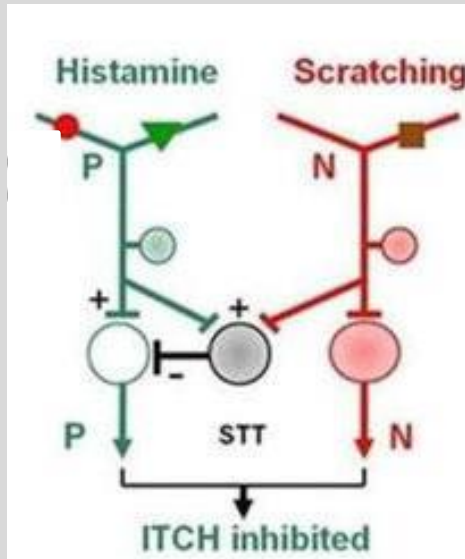
Itch (Pruritis)

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- Different behaviors for itch (scratch) versus pain (withdrawal)
- Specific to skin (no visceral itch)
- Evoked by many conditions (dry skin, pregnancy, morphine)
- Conducted by **c-fibers**

Histamine-sensitive & mechanically insensitive pathway
Histamine-independent & mechanically sensitive



Pain inhibits itch
(Similar principles as
“gate theory”)

Fill in the blank

Summary of sensory receptors

Receptor type	Characteristics	Fiber type	Conduc tion
Touch (mechanoreceptors)			
Pain (nociceptors)			
Proprioception (proprioceptors)			

Pause
here...

Summary of sensory receptors

Receptor type	Characteristics	Fiber type	Conduction
Touch (mechanoreceptor) Meissner's corpuscles Merkel cells Pacinian corpuscles Ruffini endings	Light touch, fast adapting, small RF Static touch, slow adapting, small RF Vibration & pressure, fast adapting, large RF Pressure, slow adapting, large RF	A β , large diameter, myelinated	Fast
Pain (nociceptors) Free nerve endings Noxious stimuli (thermal, chemical, etc.)	Sharp pain Dull pain	A δ , small diameter, myelinated C, small diameter, unmyelinated	Fast Slow
Proprioception Free nerve endings Joint receptors Golgi tendon organ	Extreme limb position or directions Pulling force at the junction of nerve endings and tendon collagen fibers	A α , large diameter, myelinated	Fast

Sample question #1

Stimulating the afferent for a Meissner's receptor embedded in the hand should lead to what sensation?

- A. Joint motion
- B. Sharp pain
- C. Light touch
- D. Temperature
- E. Itch

Pause here...

Sample question #1

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- E. Itch

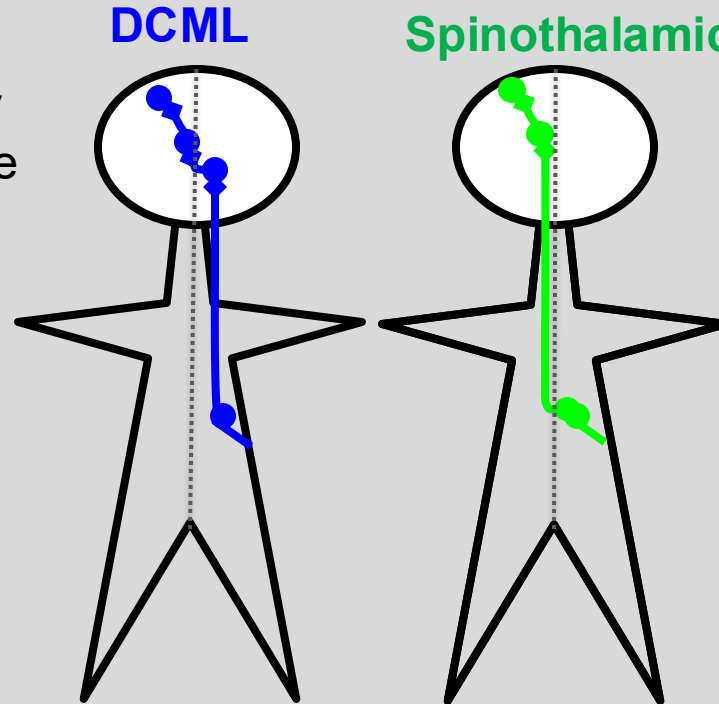
Some signposts on the road to the brain

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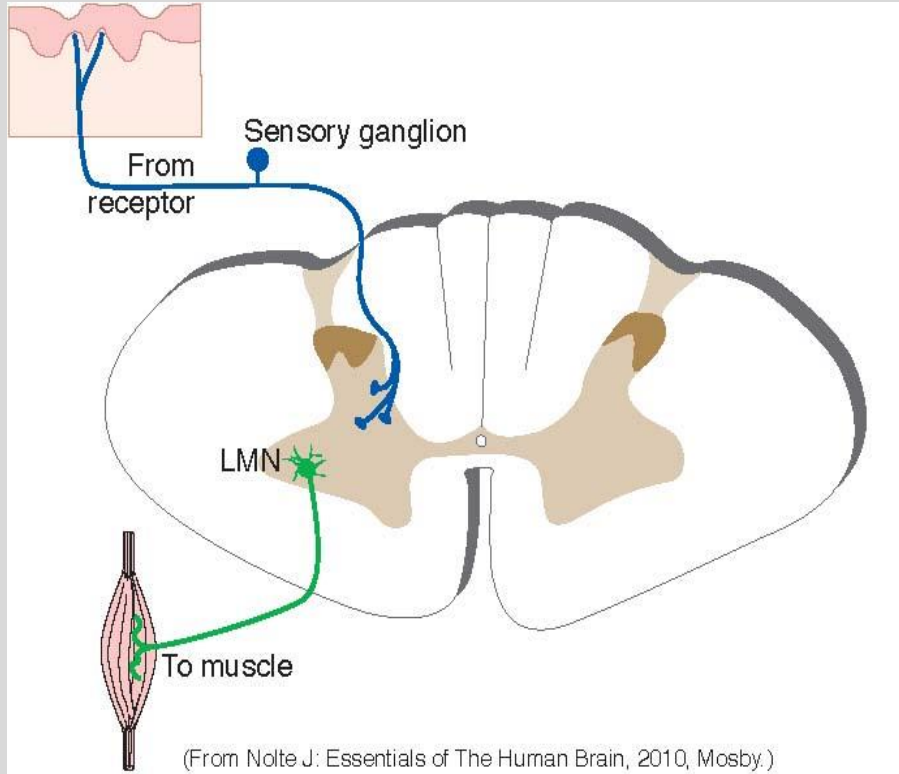
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3-neuron-relay

- 1st order neurons are sensory neurons with cell bodies in the dorsal root ganglia (DRG).
- Sensory neurons do not decussate.
- 2nd order neurons decussate.
- 3rd order neurons at the thalamus, the gateway to cortex.
- Final destination at the somatosensory cortex.



Primary Afferents



- Cell bodies of 1° afferents live outside the CNS* at the **dorsal root ganglia (DRG)** or **trigeminal ganglia (pseudounipolar neurons)**
- Primary afferents for somatic neurons end on the **ipsilateral** side of the CNS*
- Damage to 1° afferents – lose part or all **perception** and **reflex** (if one exists)

* With some exceptions

Anatomical Organization of the Nervous System

- 2° Afferents for pain and temperature cross at the level they enter the spinal cord

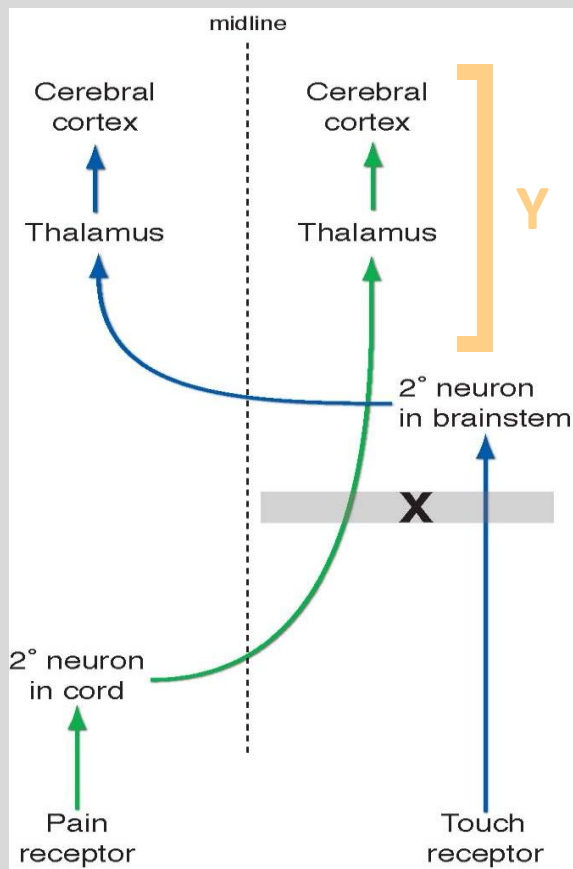
- 2° Afferents for touch and proprioception cross at the level of the brainstem

Damage at X

- **Ipsilateral** loss of touch and proprioception
- **Contralateral** loss of pain and temperature

Damage at Y

- **Contralateral** loss of both sensory modalities



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Spinothalamic Pathway

4° - Primary

sensory
cortex

3° - Ventral
posterior lateral
(VPL) thalamic
nucleus

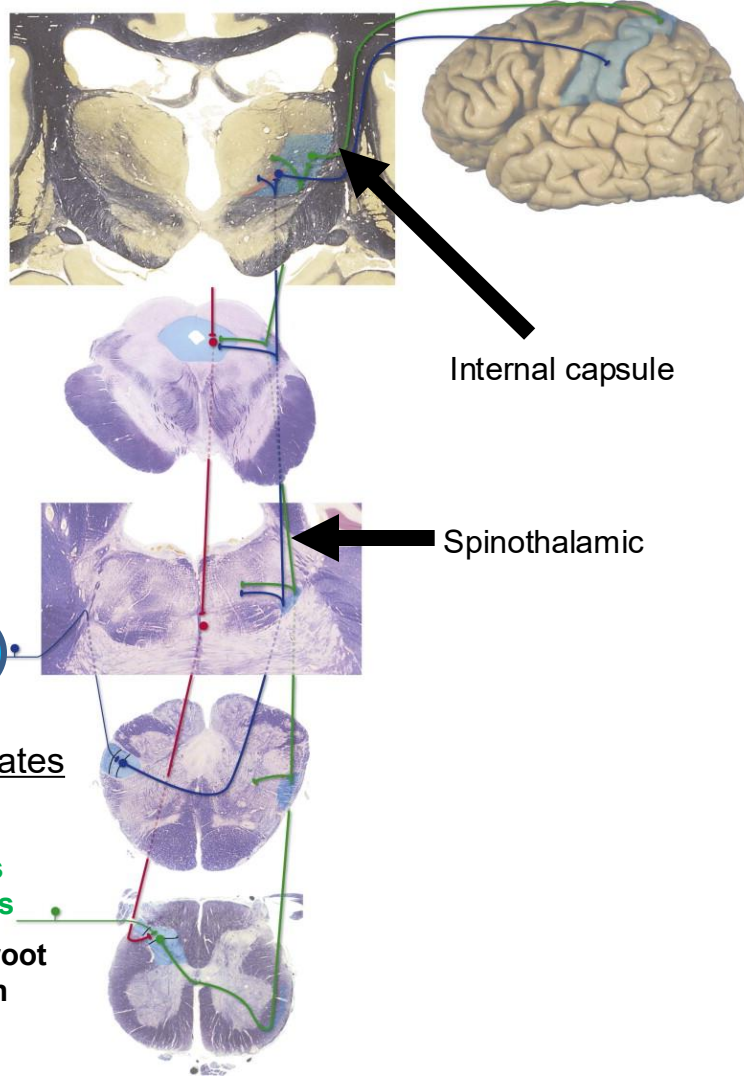
Targets primary somatosensory
cortex just like DCML

S1 processes conscious
intensity and location of
Pain, Itch, and Temperature,
not the affective experience

Spinothalamic decussates
in spinal cord

2° - Nucleus
proprius

1° - Dorsal root
ganglion



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Dorsal Column-Medial Lemniscus Pathway (DCML)

4° - Primary
somatosensor
y
cortex

3° - Ventral posterior lateral
(VPL) thalamic nucleus

Conveys "conscious proprioception"
and "discriminative touch"
Above & below T₆

Decussates at medulla

2° - Nucleus gracilis

2° - Nucleus cuneatus

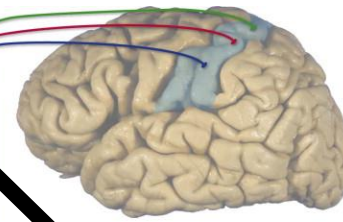
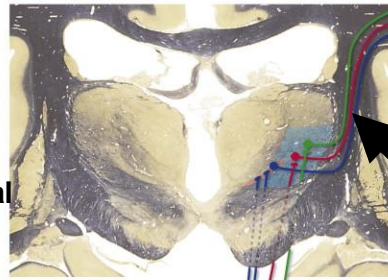
1° - Dorsal root
ganglion

1° - Dorsal root
ganglion

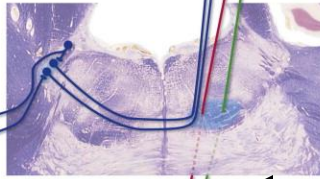
1° = 1st order in series

2° = 2nd order in series

3° = 3rd order in series



Internal capsule

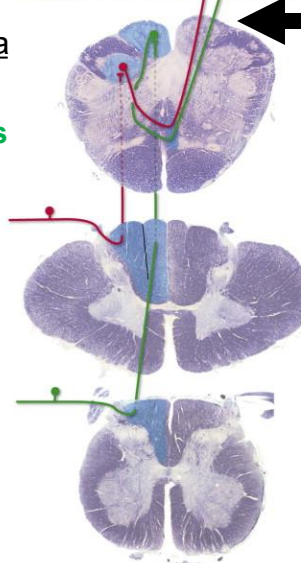


Medial lemniscus

Dorsal columns

+ Fasciculus Cuneatus
(upper half of body)

Fasciculus Gracilis
(lower half of body)



Somatosensation in the head processed by three nuclei

Principal/main/pontine trigeminal nucleus (pons)

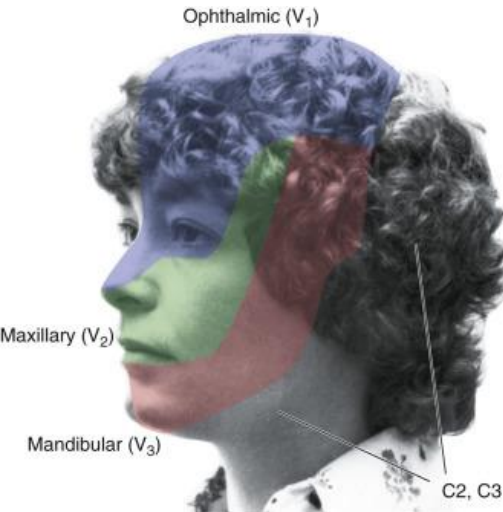
- analogous to **dorsal column nuclei** for DCML
- receives **touch and proprioceptive** information from face, cornea, tongue, ear, oronasal mucous membranes, dura, oropharynx, and laryngopharynx

Spinal trigeminal nucleus (medulla)

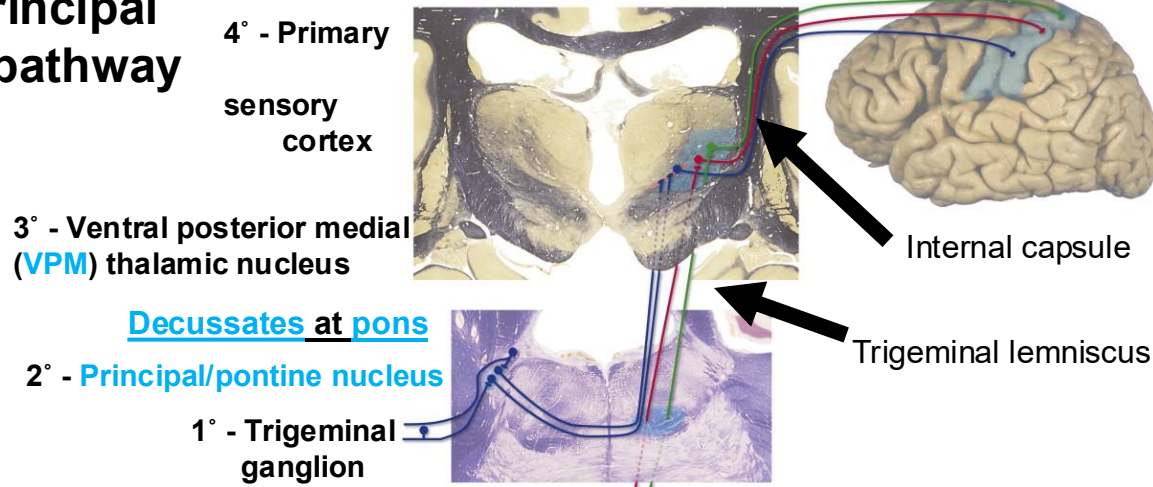
- analogous to **nucleus proprius** for STT
- receives **nociceptive** signals and **temperature** info from the same areas as the above

Mesencephalic trigeminal nucleus (midbrain)

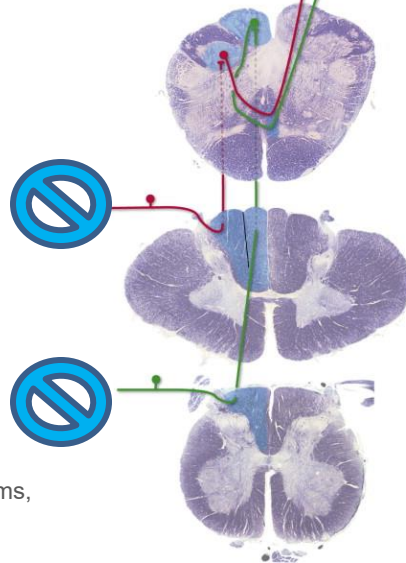
- Analogous to dorsal root ganglia (**DRG**), where the cell bodies of pseudounipolar neurons (**1st order** neurons) for the **proprioception** of jaw muscles and periodontal regions are located
- Receives proprioceptive information from the mandible and projects to the **trigeminal motor nucleus** to mediate **jaw jerk reflex**



Pontine/principal trigeminal pathway



- Conveys “conscious proprioception” and “discriminative touch” of the head
- analogous to DCML



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Spinal trigeminal pathway

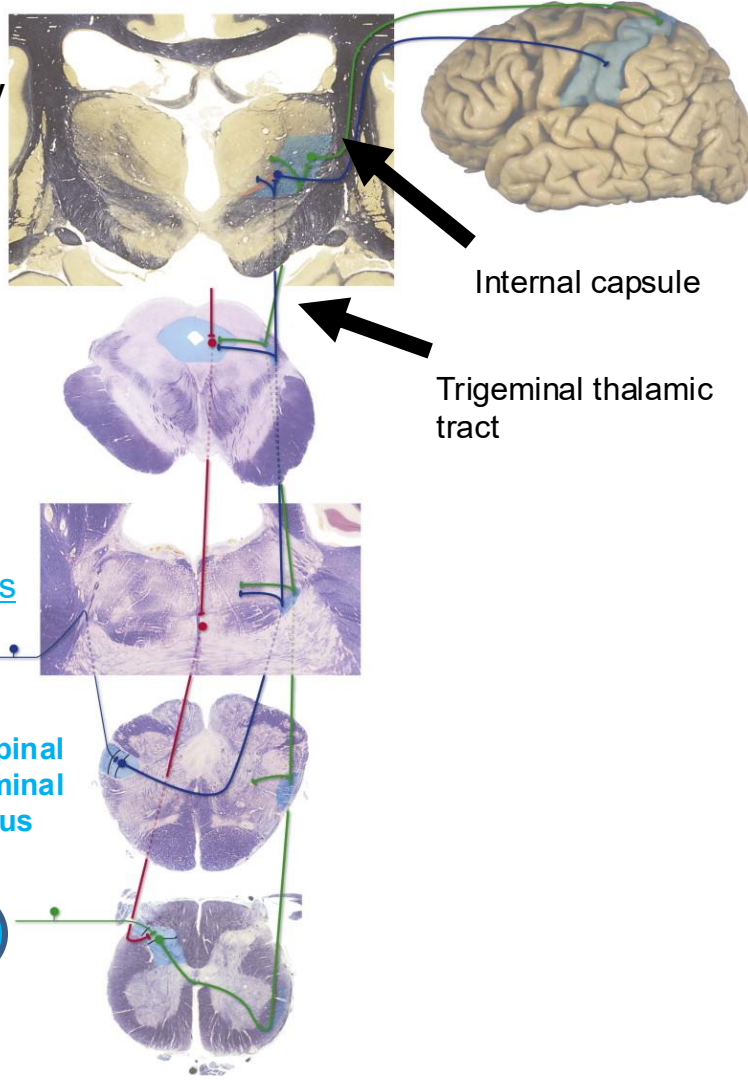
- 4° - Primary somatosensory cortex
- 3° - Ventral posterior medial (VPM) thalamic nucleus

- Conveys conscious intensity and location of **pain** (and Itch) and **temperature** information of the **head**.
- Analogous to spinothalamic tract.

Descends to medulla,
synapse and decussates

- 1° - Trigeminal ganglion

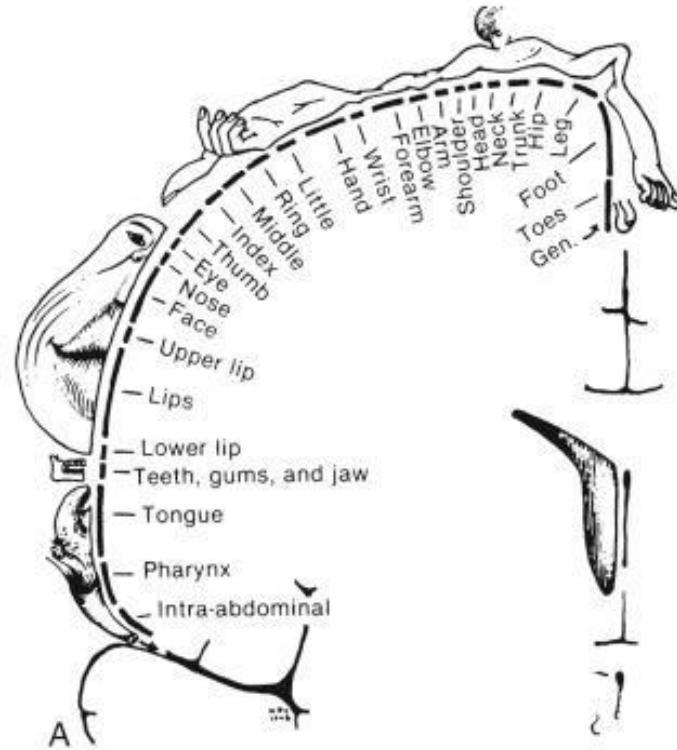
- 2° - **Spinal trigeminal nucleus**



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Sensory homunculus of the somatosensory cortex



Penfield and Rasmussen (1950)
The cerebral cortex of man

Primary somatosensory cortex is the **entire post-central gyrus including medial wall.**

Classically, this is the first receiving station within cortex for somatosensory information. Unimodal (no visual or auditory processing which occurs in more posterior parietal cortex).

“Somatotopy” – nearby portions of the body are represented by nearby portions of cortex.

Area of cortex for a given portion of the body/face reflects the density of sensory innervation.

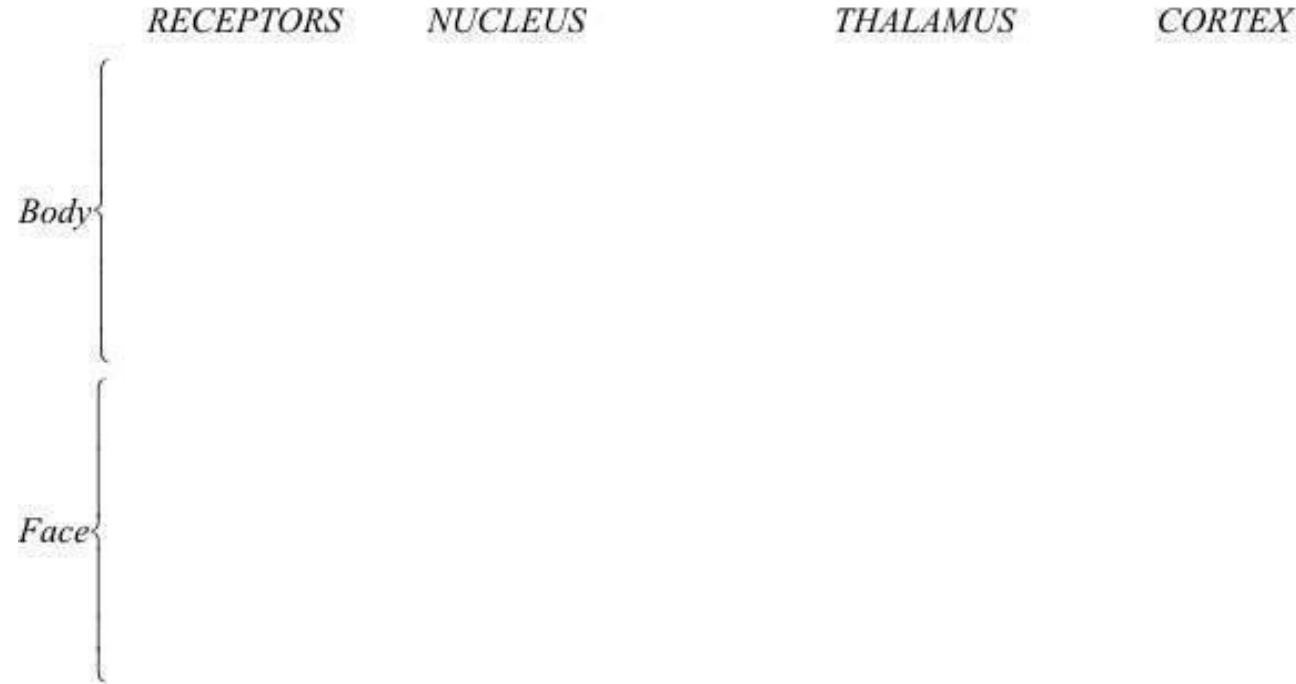
The **leg** is most **medial**, then trunk, **hand**, **face**, and then oral/pharynx.

Somatosensory pathways from the body and face

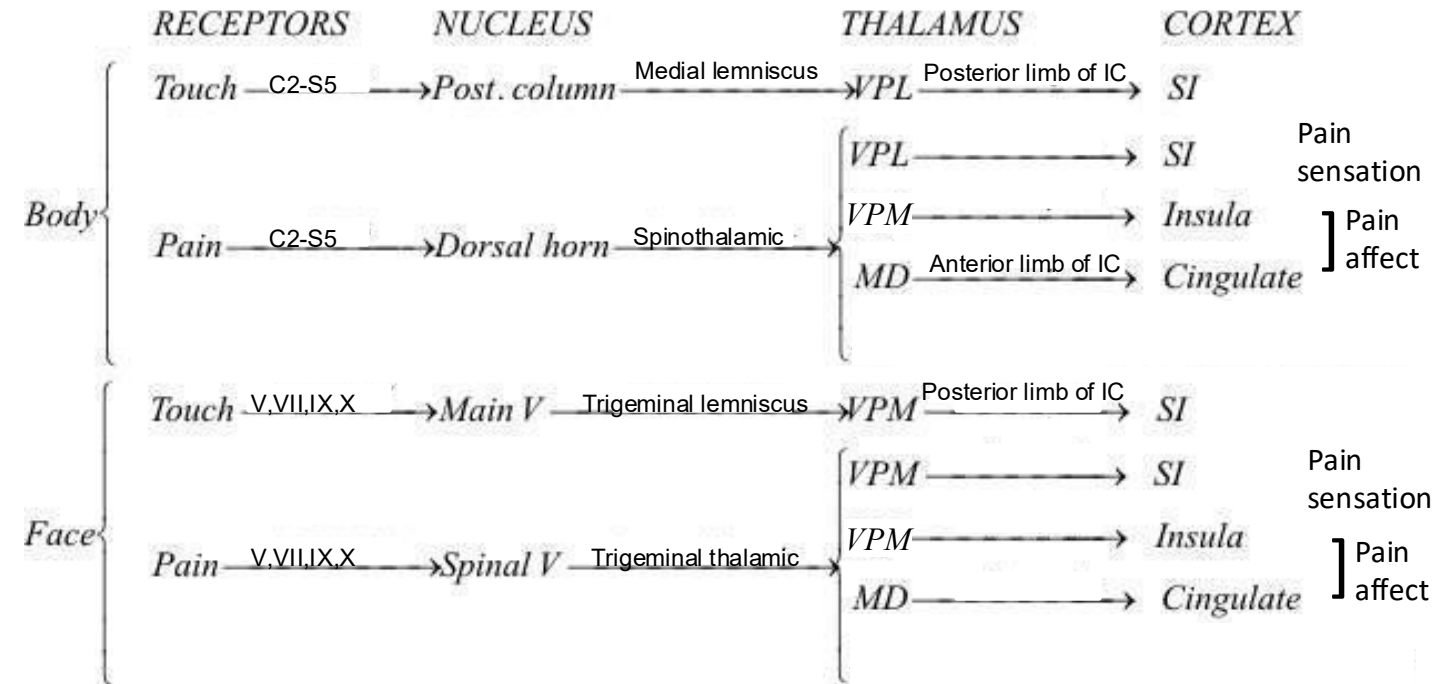
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Pause here...



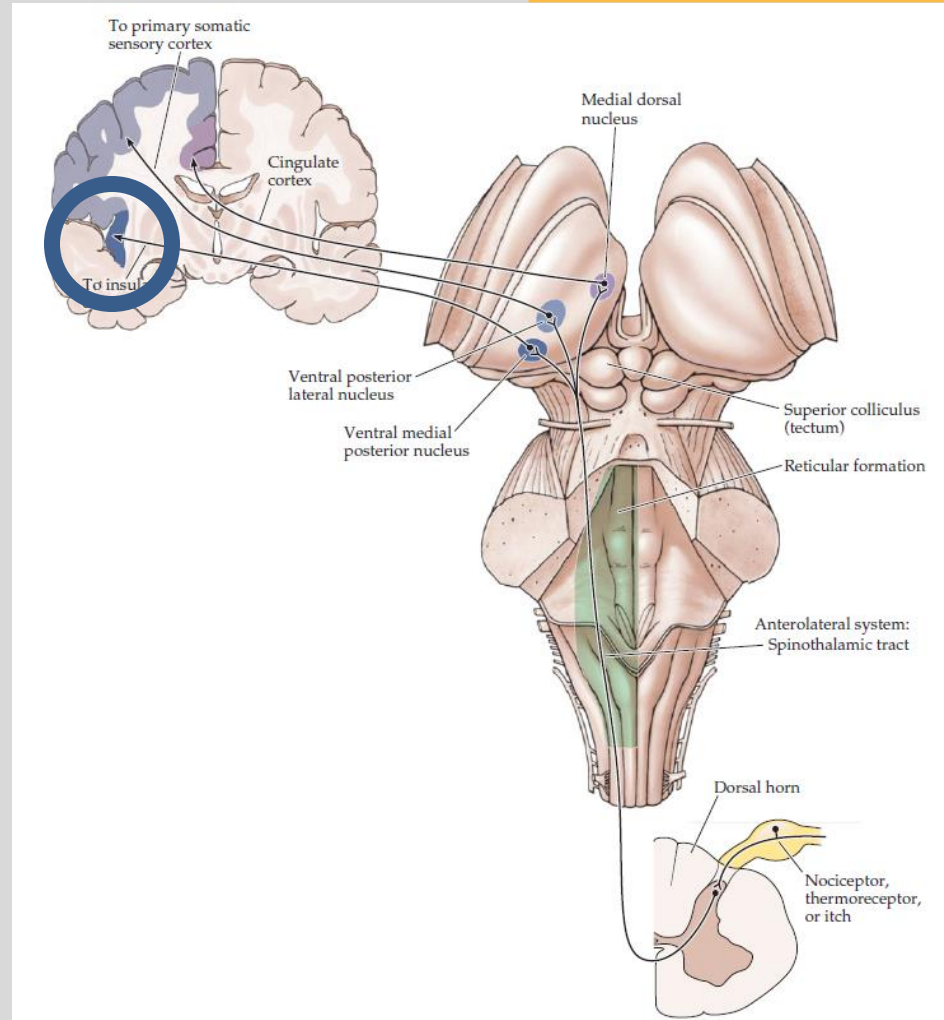
Somatosensory pathways from the body and face



“Where does it hurt?”

Ascending nociceptive pathway also projects to **insular** cortex and **cingulate** cortex, which process the **affective** (unpleasantness) aspects of pain.

For example, dense lesion of insula can lead to blunted pain thresholds for some stimuli despite the ability to recognize and rate intensity.



'Phantoms' of the body

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The compelling experience that the body part is still present following its **sudden removal** whether by surgery, accident, or violence.

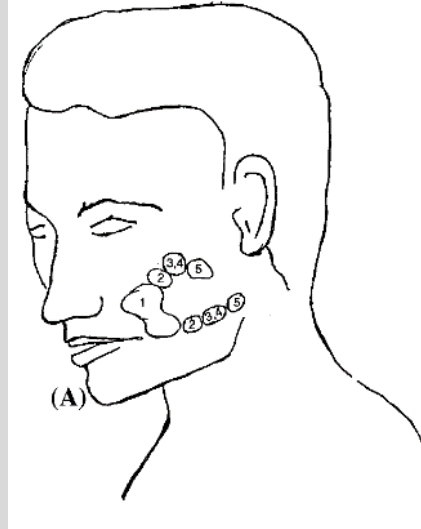
Phantom arm, phantom leg, ...

Often painful sensation (e.g. burning, fingernails pressing into skin, contortions) and typically weak voluntary control. Resistant to medication.

Body representation changes but still orderly.

Cortical reorganization as face area expands into unused arm area.

Ramachandran and Hirstein (1990)
The perception of phantom limbs



Ipsilateral or contralateral?

- Lesions **above** the decussation cause **contralateral** deficits;
below the decussation cause **ipsilateral** deficits

What kind of deficits would be presented when the lesions are at these levels?

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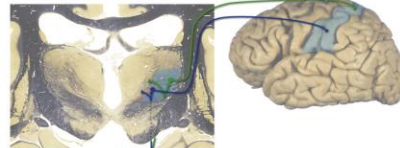
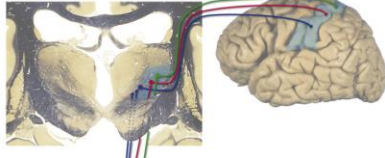
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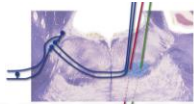
Touch/Proprioception

Pain/Temperature

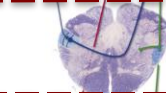
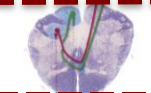
Forebrain



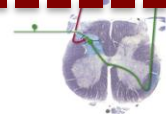
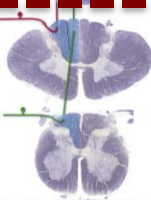
Pons



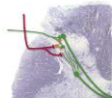
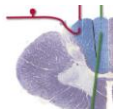
Medulla



Spinal
cord



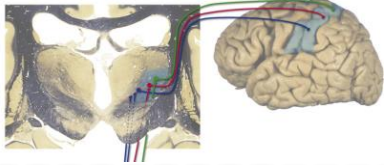
Primary
afferent



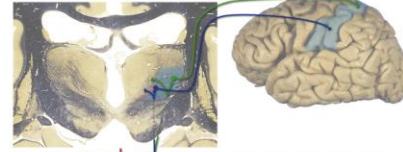
What kind of deficits at different levels?

Touch/Proprioception

Pain/Temperature



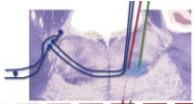
Contralateral
Face & Body



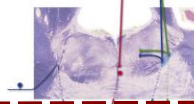
Contralateral
Face & Body

Forebrain

Pons

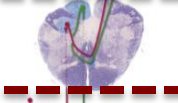


Ipsi-/Contra-lateral face;
Contralateral body



Ipsi-/Contra-lateral face;
Contralateral body

Medulla

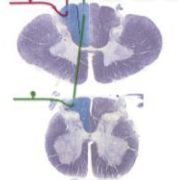


Contralateral body

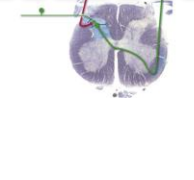


Ipsi-/Contra-lateral face;
Contralateral body

Spinal
cord

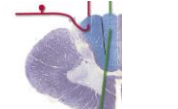


Ipsilateral body

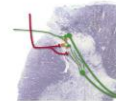


Contralateral body

Primary
afferent



Ipsilateral dermatome



Ipsilateral dermatome

Test yourself

Summary

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Lesions at the **spinal** levels –

_____ deficits for **pain/temperature** sensation

_____ deficits for **touch/proprioception** sensation
and **voluntary movement** for the body

Lesions **above brainstem** (e.g., thalamus, internal capsule, cortex) -

_____ deficits for **all modalities** for the body

Lesions **at brainstem** (e.g., midbrain, pons, medulla) -

_____ deficits for **the body**;

_____ deficits for **the face**

Pause
here...

Summary

Lesions at the **spinal** levels –

contralateral deficits for **pain/temperature** sensation

ipsilateral deficits for **touch/proprioception** sensation
and **voluntary movement** for the body

Lesions **above brainstem** (e.g., thalamus, internal capsule, cortex) -

contralateral deficits for **all modalities** for the body

Lesions **at brainstem** (e.g., midbrain, pons, medulla) -

contralateral deficits for **the body**;

ipsi- or contra-lateral deficits for **the face**
(depending on the level of the CN nuclei)

Sample question #2

A lesion to the right ventral posterolateral nucleus would blunt this sensation:

- A. Touch and proprioception of the right face
- B. Touch and proprioception of the left face
- C. Touch and proprioception to the left face and body
- D. Touch and proprioception to the right body
- E. Touch and proprioception to the left body

Pause
here...

Sample question #2

A lesion to the right ventral posterolateral nucleus would blunt this sensation:

- A. Touch and proprioception of the right face
- B. Touch and proprioception of the left face
- C. Touch and proprioception to the left face and body
- D. Touch and proprioception to the right body
- E. Touch and proprioception to the left body

Sample question #3

A 76-year-old male presents to the emergency department with complaint of fatigue and general weakness. History reveals hypertension. Physical examination reveals hemiparesis in his upper and lower extremities on the left side, as well as hemianesthesia for the left side of the body and face. Visual field examination is unremarkable. Which of the following is the most likely location for a stroke causing these clinical signs?

- a) Genu of internal capsule
- b) Lateral geniculate nucleus of thalamus
- c) Posterior limb of internal capsule
- d) Ventral posterolateral nucleus of thalamus
- e) Ventral posteromedial nucleus of thalamus

Pause
here...

Sample question #3

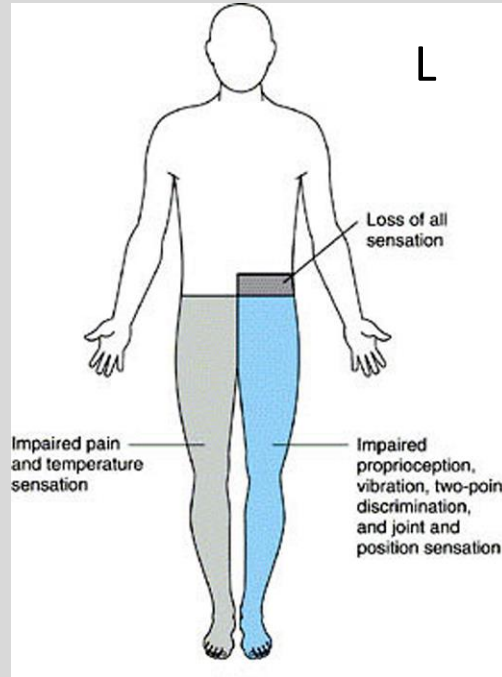
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- a) Genu of internal capsule
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Find the lesion

At what level?

Which side?



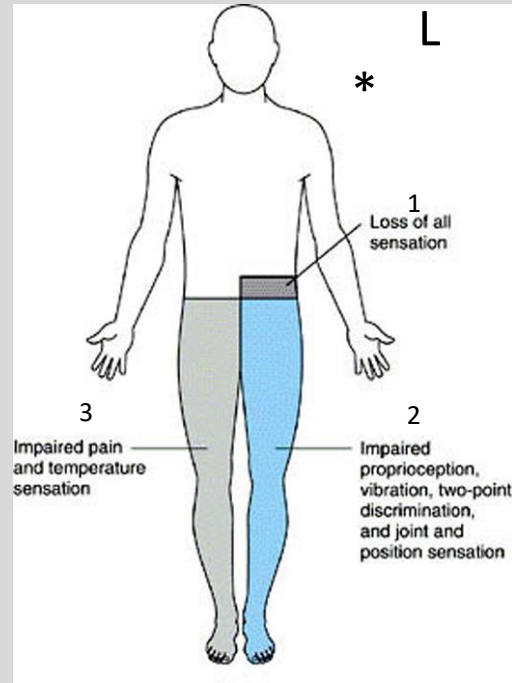
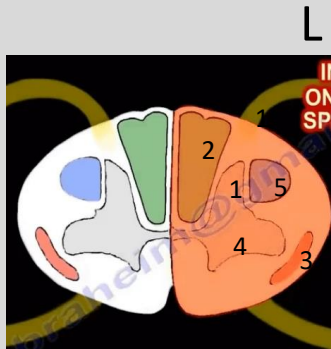
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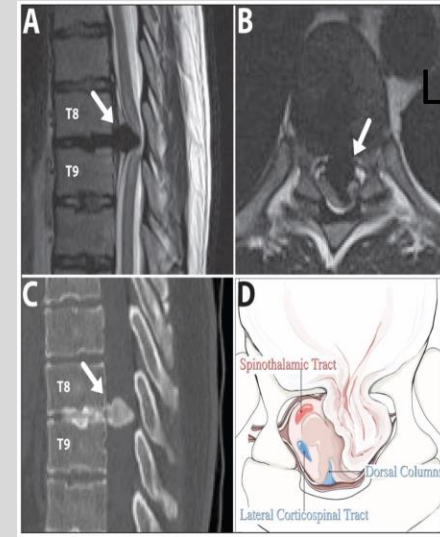
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Brown-Sequard syndrome

At what level?
Which side?



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All lesions at the lower body – damage to the lumbar cord:

1. loss of all sensation **at the level** of the lesion at the left side – **ipsilateral dorsal horn** lesion
2. loss of proprioception, vibration, discriminative touch, and tactile sense **below the level** of the lesion at the left side – **ipsilateral DCML** lesion
3. loss of pain, temp, and crude touch **below the level** of the lesion at the right side – **contralateral spinothalamic tract** lesion
4. LMN signs **at the level** of the lesion at the left side – **ipsilateral ventral horn** lesion
5. UMN signs **below the level** of lesion at the left side – **ipsilateral corticospinal tract** lesion

Lecture Feedback Form _ Dr. Jennifer Xie

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>



Email: jennifer.xie@nyit.edu